

Grade 9 Readiness Screen

Student Booklet - Version 2.0

Student name	
Date of birth	
Assessment date	
Parent/guardian	
Current city and country	

This short screen helps GIIS identify a responsible starting point. It does not by itself award credit, guarantee Grade 9 placement, or create a transcript grade.

Return completed pages to: admissions@genesisideas.school

Family and Student Instructions

About 90 minutes total. It may be completed in one sitting or two shorter sessions.

1 Print

Print the full booklet at 100% scale. One-sided printing is recommended.

2 Work independently

A parent may explain procedural directions, but may not translate passages, suggest methods, check answers, or edit the work.

3 Use only allowed tools

English: no dictionary, translator, AI, or internet. Math Questions 1-6: no calculator. Math Questions 7-12 and Science: a basic calculator is allowed.

4 Write and show work

Write by hand on the booklet. Attach labeled extra paper if needed. If you do not know an answer, write 'I do not know' and continue.

5 Return clear pages

Scan everything into one PDF if possible. Clear, flat, well-lit photos are also acceptable.

Suggested timing

Section	Time	Focus
English	35 min	Reading and short writing
Mathematics	35 min	Core Grade 8-to-9 prerequisites
Science reasoning	20 min	Data, evidence, and investigation
Learning check	Untimed	Planning and independent work habits

The goal is an honest snapshot, not a perfect score. Take a short break between sections if needed.

English

30 points | 35 minutes | No dictionary, translator, AI, or internet

A Cooler Route to School

On warm afternoons, students walking home through Riverton's East Ward often waited for buses on blocks with little shade. A student group wanted the city to plant trees, but they first needed evidence that the hottest locations were also places students regularly used.

The group selected six bus stops and recorded surface temperature, air temperature, shade, and the number of waiting passengers at the same time on four clear days. The pavement at the least shaded stops was consistently hotter. Two of those stops also had the highest passenger counts.

The students proposed planting trees at all six stops. A city planner supported the goal but explained that young trees require water and care for years before they provide much shade. Without a maintenance plan, some trees might die before helping anyone.

The students revised their proposal. They focused on the two busy, high-temperature stops and found community groups willing to water the trees. They also recommended temporary shade structures while the trees grew. The revised plan covered fewer locations, but it connected each recommendation to evidence and a realistic maintenance partner.

The project did not prove that trees solve every danger caused by heat. It did show how measurements and feedback can turn a broad concern into a practical first step.

1. State the central idea. Give two details from the passage that support it. [6 points]

2. Why does the author include the city planner's concern? Explain how it changes the students' proposal. [4 points]

Bus stop	Average pavement temperature	Average waiting passengers
A - full shade	91 F	8
B - partial shade	102 F	11
C - little shade	112 F	19

SECTION 2A

Mathematics - No Calculator

18 points | About 17 minutes | Show work

1. Evaluate: $-5 + 3(4 - 7)$. [2 points]

2. Compute and simplify: $\frac{3}{4} + \frac{5}{8}$. [3 points]

3. A \$64 jacket is discounted 25%. What is the sale price? [3 points]

4. A recipe uses 5 cups of flour for 12 servings. How many cups are needed for 30 servings? [3 points]

5. Identify the two consecutive integers between which the square root of 50 lies. Explain. [3 points]

6. Solve and check: $3(2x - 5) = 4x + 7$. [4 points]

Mathematics - Basic Calculator Allowed

18 points | About 18 minutes | Show equations, reasoning, and units

7. A line passes through (1, 4) and (5, 10). Find its slope. [3 points]

Use the table for Questions 8-9.

x	-1	0	1	2
y	7	4	1	-2

8. Is the relationship linear? Write an equation for y in terms of x. [4 points]

9. Compare the signed rate of change in the table with $g(x) = 2x - 1$. Which function increases as x increases, and which decreases? [3 points]

10. A right triangle has legs 9 cm and 12 cm. Find the hypotenuse and name the theorem used. [3 points]

11. Plan A costs \$12 plus \$2 per movie. Plan B costs \$24 plus \$0.50 per movie. Write an equation for each plan and find when the costs are equal. [3 points]

12. A student says the only solution to $2(x + 4) = 2x + 8$ is $x = 0$. Is the student correct? Explain. [2 points]

SECTION 3

Science Reasoning

24 points | 20 minutes | Basic calculator allowed | Use evidence

Investigation 1 - Rolling Cart

Surface	Trial 1 distance	Trial 2 distance	Trial 3 distance
Smooth floor	2.3 m	2.5 m	2.4 m
Carpet	0.8 m	1.0 m	0.9 m

1. The same cart was released from the same ramp onto each surface. Which surface produced more friction? Calculate the average distance on each surface, use the data as evidence, and name one other variable that should be controlled. [8 points]

Investigation 2 - Plant Growth

Group	Light/day	Fertilizer	Average growth
A	8 hours	None	6.2 cm
B	8 hours	5 g/week	9.1 cm
C	4 hours	5 g/week	4.8 cm
D	4 hours	None	3.9 cm

2. Use two comparisons from the table to explain how light and fertilizer affected growth. Give one reason this study cannot prove the same result for every plant species. [8 points]

Evidence and Design

Complete Question 3, then the unscored learning check.

Design	Keeps water below 12 C	Cost	Reusable
A	Yes	\$18	Yes
B	No	\$10	Yes
C	Yes	\$14	No

3. The required criteria are: keeps water below 12 C, costs no more than \$15, and is reusable. Does any design meet all criteria? If not, recommend one specific revision and explain its tradeoff. [8 points]

Independent Learning Check - not scored

1. ☐ Usually ☐ Sometimes ☐ Not yet I can follow a multi-step assignment and identify what to submit.
2. ☐ Usually ☐ Sometimes ☐ Not yet I can work independently for 30-45 minutes before asking for help.
3. ☐ Usually ☐ Sometimes ☐ Not yet I can name and attach the correct file.
4. ☐ Usually ☐ Sometimes ☐ Not yet I can explain which step is confusing when I need help.
5. ☐ Usually ☐ Sometimes ☐ Not yet I can follow a weekly schedule and report early if I fall behind.

Which subject felt strongest? Which skill needs the most practice?

Family Verification and Submission

Please return every completed page, including extra work.

By signing below, we confirm that the student completed the academic sections independently using only the tools listed in the instructions. Any interruption, translation, or assistance is disclosed below.

Student signature		Date	
Parent/guardian signature		Date	

Interruptions or assistance to disclose:

Return checklist

☐ All booklet pages ☐ Extra work pages ☐ Two recent independent work samples ☐ Khan Academy/IXL evidence, if available

Email to admissions@genesideas.school

Suggested subject: Grade 9 Readiness Assessment - [Student Full Name]

GIIS reviews the work by skill area and may request a short oral walkthrough. The Principal then provides a written placement decision: Ready for a conditional Grade 9 start, Ready with a named bridge plan, or Not Yet with a preparation recommendation. If required records or circumstances are unclear, the case remains Pending Clarification rather than receiving an academic result.